

BOTANY HUB MULTAI

B.Sc. I BOTANY — MINOR I

ELEMENTARY BOTANY

UNIT III — Internal Structure

DETAILED TEACHER-STYLE NOTES

Definitions • Every concept explained • Tables • Labelled study diagrams • Flowcharts • Examples • Comparisons • Exam points

Prepared in the same detailed format as Major Course II and Major Course III notes.

UNIT III — INTERNAL STRUCTURE

The supplied notes identify plant cell, tissues, and root–stem–leaf anatomy as the core of this unit.

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1. Plant cell

Definition — Plant cell: A eukaryotic cell forming the basic structural and functional unit of plant body, characterized by a plasma membrane and typically a cellulose-rich cell wall, large vacuole and, in photosynthetic tissues, chloroplasts.

Component	Main function	Exam keyword
Cell wall	Protection and shape	Cellulose-rich outer layer
Plasma membrane	Selective movement of substances	Selectively permeable
Cytoplasm	Medium for cellular reactions	Metabolic activities
Nucleus	Genetic control	DNA/chromatin
Chloroplast	Photosynthesis	Chlorophyll
Mitochondrion	ATP generation in aerobic respiration	Energy transformation
Vacuole	Storage and turgor	Cell sap

2. Plant tissues

Definition — Tissue: A group of cells with common origin and coordinated function.

Meristematic tissue

These cells retain the ability to divide. They are usually small, compact and metabolically active.

Type	Location	Main result
Apical meristem	Root and shoot tips	Primary growth / increase in length
Intercalary meristem	Often near nodes or leaf bases in grasses	Rapid elongation/regrowth
Lateral meristem	Along sides of stems/roots	Secondary growth / increase in girth

Permanent tissues

Tissue	Key structure	Function
Parenchyma	Living, relatively thin-walled cells	Storage, metabolism, photosynthesis in chlorenchyma
Collenchyma	Living cells with uneven wall thickening	Flexible support
Sclerenchyma	Usually dead at maturity with thick walls	Mechanical strength
Xylem	Complex vascular tissue	Water/mineral transport
Phloem	Complex vascular tissue	Transport of organic solutes

Functional organization of plant body



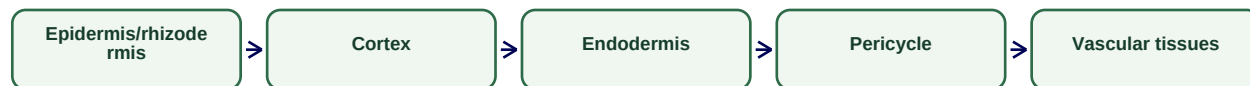
3. Basic anatomy of root, stem and leaf

Organ	Main internal adaptation	Functional advantage
Root	Absorptive outer region and central vascular tissue	Efficient absorption and upward conduction
Stem	Protective outer region, ground tissue and vascular bundles	Support and long-distance transport

Organ	Main internal adaptation	Functional advantage
Leaf	Epidermis, mesophyll and veins	Photosynthesis plus controlled gas exchange

Root anatomy—study sequence

Simplified root cross-section sequence



Leaf anatomy—study sequence

Simplified dorsiventral leaf plan



Exam focus: Draw only diagrams you can label correctly. A clean simplified cross-section with 5–7 labels scores better than an artistic but incorrect diagram.